

C A S E S T U D Y · 2 0 2 6

MitchAI

Hospitality software prototype case study

Status & implementation boundary — What this case study does and does not claim

Status: MitchAI is a technical portfolio prototype, not a public-facing production service.

Repository: <https://github.com/Bogdan0708/MitchAI> An n8n master orchestrator (published as 8 sanitised workflows in automations/n8n) exists and is checked in, and the app is deployed to AWS ECS Fargate + RDS via a GitHub Actions workflow (green run 2026-02-12, image build + ECS service update); Terraform modules for GCP and AWS are published separately and are not shown to have provisioned that run. Neither is offered as a hosted product with an SLA, and no independently assessed GDPR/compliance status is claimed.

Role: Founder and engineer, drawing on 10+ years of hospitality operations.

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01 Product scope

Hospitality workflows represented in code

Checked-in scope

- Routes and services for menus, orders, reservations, reviews, content, analytics and AI-assisted tasks.
- n8n master orchestrator routing to Menu / Social / Strategy / Feedback / Operations workflows (automations/n8n, 8 sanitised workflows).
- HACCP compliance module; WhatsApp Cloud API and Telegram agent; 53 row-level security policies.
- Authentication, tenant context, validation, rate limiting and PostgreSQL schema/migrations; a Next.js dashboard for hospitality and AI workflow surfaces.
- 7-provider LLM orchestrator, including local Ollama and LM Studio alongside cloud providers.

Boundary

Modules may require credentials, external services and further integration testing. They are implementation evidence, not evidence that every workflow runs end to end for customers.

02 Architecture

Primary runtime stated accurately

Primary stack

The backend is Node.js with TypeScript and Express. The frontend is Next.js/React, and PostgreSQL — behind 53 row-level security policies — is the primary persistence layer. Redis and Qdrant are supporting services.

n8n is MitchAI's production master orchestrator (published workflows), not an optional add-on; Ollama and LM Studio are the local-model options inside the 7-provider orchestrator. Python and FastAPI are not the primary MitchAI runtime.

Deployed to AWS ECS Fargate + RDS via a GitHub Actions workflow (green run 2026-02-12, image build + ECS service update); Terraform modules for GCP and AWS are published separately and are not shown to have provisioned that run; CI green.

Engineering posture

- Human approval remains the intended boundary for posts, messages, filings and purchases.
- Security-oriented controls (row-level security policies, HACCP module) are code-level signals, not certification or a live-service guarantee.

03 Evidence & next steps

What the prototype demonstrates

What it proves

MitchAI demonstrates a broad hospitality-domain TypeScript/Express backend, a published n8n orchestration layer, a Next.js frontend, a row-level-secured PostgreSQL data model, a 7-provider AI orchestrator, a GitHub Actions-driven AWS ECS Fargate + RDS deployment (green run 2026-02-12; Terraform modules published separately, not shown to have provisioned that run), and CI/test configuration.

Remaining gates

- Validate complete workflows against configured local dependencies.
- Publish current test/build outputs rather than fixed or historical totals.
- Complete security, privacy and deployment review before production or compliance language.